How Costly Are Markups?

An Analysis of Edmond, Midrigan, and Xu (2023)

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Introduction & Research Question

- **Core Question:** What are the aggregate welfare costs of markup distortions?
- **Three Channels of Distortion:**
  1. **Aggregate Markup:** Acts as a uniform output tax $\rightarrow$ lowers economic scale.
  2. **Misallocation:** Markup dispersion $\rightarrow$ productive firms under-produce.
  3. **Inefficient Entry:** Misalignment between private and social incentives.

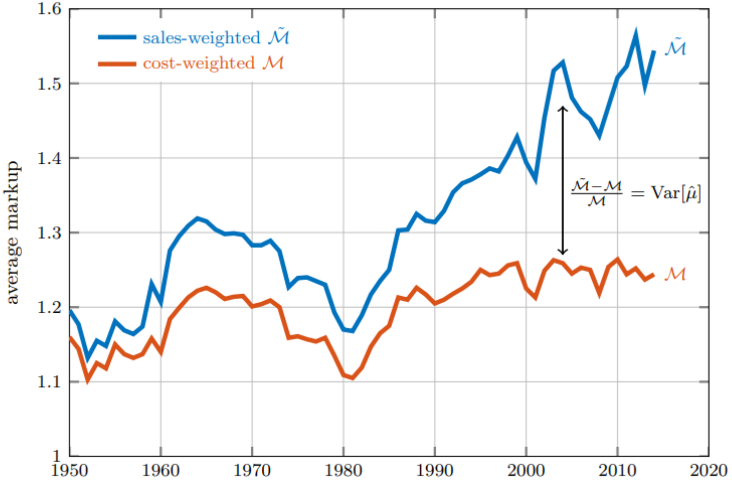
Methodological Contribution: Cost vs. Sales Weights

- **Empirical Literature:** Uses *sales-weighted* average markups ($\tilde{\mathcal{M}}$).
- **Macro Theory:** True resource wedge is the *cost-weighted* average ($\mathcal{M}$).
- **The Analytical Relationship:**

$$\frac{\tilde{\mathcal{M}} - \mathcal{M}}{\mathcal{M}} = Var[\hat{\mu}_i]$$

- **Insight:** Sales-weighted metrics overstate market power by capturing markup variance (misallocation).

Methodological Contribution: Cost vs. Sales Weights



Model Setup: Households & Final Good

- **Representative Consumer:**

$$\sum_{t=0}^{\infty} \beta^t \left(\log C_t - \psi \frac{L_t^{1+\nu}}{1+\nu} \right)$$

- **Final Good Sector :**

$$Y_t = \left(\int_0^1 y_t(s)^{\frac{\eta-1}{\eta}} ds \right)^{\frac{\eta}{\eta-1}}$$

- **Roundabout Structure:**

$$C_t + I_t + X_t = Y_t$$

- Final good is used for consumption, investment, and intermediate materials (X_t).

Intermediate Goods Production

- **Firms:** Heterogeneous productivity $z_i(s) \sim \text{Pareto}(\xi)$.
- **Nested Production Function:**

$$y_{it}(s) = z_i(s) \left(\phi^{\frac{1}{\theta}} v_{it}(s)^{\frac{\theta-1}{\theta}} + (1-\phi)^{\frac{1}{\theta}} x_{it}(s)^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}$$

- **Value-Added Component:** :

$$v_{it}(s) = k_{it}(s)^{\alpha} l_{it}(s)^{1-\alpha}$$

- **Result:** Common input ratios $\rightarrow$ firm marginal cost is simply $\frac{\Omega_t}{z_i(s)}$.

Pricing & Endogenous Markups

- **Pricing Rule:** Price is a markup over marginal cost:

$$p_{it}(s) = \mu_{it}(s)\frac{\Omega_t}{z_i(s)}, \quad \mu_{it}(s) = \frac{\sigma_{it}(s)}{\sigma_{it}(s) - 1}$$

- **Kimball Demand (Klenow & Willis, 2016):** Log-linear elasticity:

$$\sigma(q) = \overline{\sigma} q^{-\epsilon/\overline{\sigma}}$$

- **Super-elasticity ($\epsilon/\overline{\sigma} > 0$):**
  - Larger firms face less elastic demand.
  - Larger, more productive firms charge systematically higher markups.

Free Entry Condition

- **Sunk Entry Cost:** Paid in units of labor (κW_t).
- **Equilibrium Entry:** Driven by ex-ante expected discounted profits:

$$\kappa W_t = \beta \sum_{j=1}^{\infty} (\beta(1-\varphi))^{j-1} \frac{C_t}{C_{t+j}} \int_0^1 \overline{\pi}_{t+j}(s) ds$$

- **Inefficiency:**
  - Private entry depends on individual markups/profits.
  - Social planner values aggregate productivity elasticity from variety.

Quantitative Strategy & Calibration

- **Data Source:** US Census of Manufactures (1972–2012).
- **Key Targets:**
  - **Sales Concentration:** Top 5% firms = 57
  - **Markup-Size Slope:** Empirical regression $b = 0.16 \rightarrow$ Pins down $\epsilon/\overline{\sigma}$.
  - **Materials Share:** Target of 45
- **Approach:** Model evaluated across various aggregate markup targets ($\mathcal{M} = 1.05$ to 1.25).

Quantitative Strategy & Calibration

Panel A: Assigned Parameters

β	discount factor	0.96
δ	depreciation rate	0.06
φ	exit rate	0.04
α	elasticity of value-added to capital	1/3
ν	elasticity of labor supply	1
θ	elasticity of substitution between value-added and materials	0.5

Panel B: Calibrated Parameters

			low	medium	high
ξ	Pareto tail		20.70	6.84	4.07
$\bar{\sigma}$	demand elasticity		29.10	10.86	7.21
$\varepsilon/\bar{\sigma}$	super-elasticity		0.16	0.16	0.16
ϕ	weight on value-added		0.51	0.43	0.33
calibration targets		*data*			
$\mathcal{M}$	aggregate markup		1.05	1.15	1.25
	top 5% sales share	0.57	0.57	0.57	0.57
	materials share	0.45	0.45	0.45	0.45
$\hat{b}$	regression coefficient	0.16	0.16	0.16	0.16

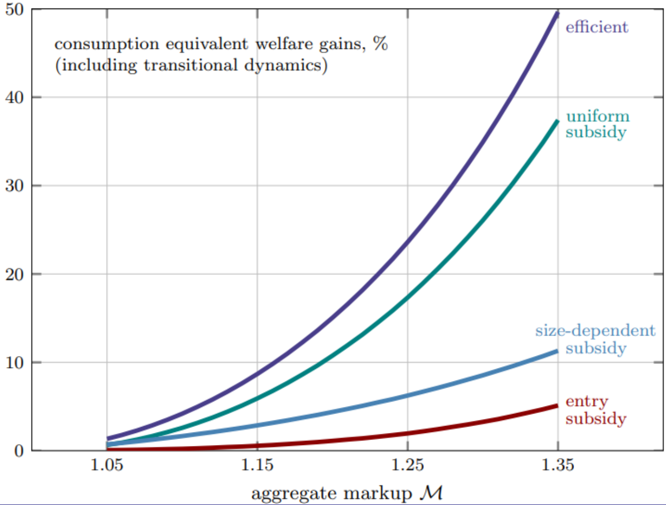
Main Results: Total Welfare Costs

- **Core Metric:** Consumption-equivalent welfare gain from removing all distortions.
- **Baseline Results ($\mathcal{M} = 1.15$):**
  - Total Welfare Gain: **8.69%**.
  - Gross Output (Y): **+59.7%**.
  - Capital Stock (K): **+100.6%**.
- **High Target ($\mathcal{M} = 1.25$):** Welfare cost jumps to **23.62%**.

Decomposing the Channels (Policy Experiments)

		steady state comparisons, %						
		Y	C	L	N	K	Z	welfare, %
$\mathcal{M} = 1.05$	efficient	15.3	11.0	6.0	12.3	24.1	0.9	1.35
	uniform subsidy	13.7	9.1	5.7	3.4	22.0	0.2	0.65
	size-dependent subsidy	1.4	1.7	0.3	8.1	1.8	0.7	0.72
	entry subsidy	1.1	1.3	0.4	11.2	1.4	0.6	0.07
$\mathcal{M} = 1.15$	efficient	59.7	44.6	18.0	20.1	100.6	4.1	8.69
	uniform subsidy	51.9	35.9	17.0	9.5	88.7	1.5	5.92
	size-dependent subsidy	5.3	6.2	1.0	8.3	6.6	2.3	2.87
	entry subsidy	6.3	7.5	2.4	20.1	8.2	3.1	0.56
$\mathcal{M} = 1.25$	efficient	134.0	102.3	30.1	26.7	246.1	8.9	23.62
	uniform subsidy	112.7	79.7	28.2	15.0	208.2	3.9	17.34
	size-dependent subsidy	10.8	12.5	1.8	8.1	13.6	4.1	6.25
	entry subsidy	17.4	20.4	6.1	29.3	23.0	7.4	1.96

Main Results: Total Welfare Costs



Why Don't Entry Subsidies Work?

- **The Paradox:** Entry sparks competition, yet aggregate markup $\mathcal{M}$ barely drops.
- **The Compositional Effect :**
  - Small, low-markup firms have elastic demand $\rightarrow$ contract heavily under entry.
  - Large, high-markup firms have inelastic demand $\rightarrow$ insulated from entry.
- **Net Result:** Reallocation gives more weight to large, high-markup firms $\rightarrow$ offsets the direct effect.

Conclusion

- Markup distortions generate heavy macroeconomic losses (up to 25
- roundabout structure via material inputs amplifies micro distortions into macro wedges.
- Pro-competitive entry policies are too blunt due to compositional effects.
- **Policy Focus:** Direct uniform production subsidies are far more effective.
- **Measurement:** Aggregate wedges must be cost-weighted, not sales-weighted.